

Lesson 1: Project - Innovation Simulation Part 1

90 minutes

Overview

The class begins a simulation which will continue on at different points throughout the unit. In this simulation, students take on the roles of different stakeholders in school communities converging at a convention where they eventually will deliver a proposal on the best computing innovation for the Future School.

On the first two day, students explore what a computing innovation is, do a brainstorm activity, reflect on their character role, and finally learn how to research an innovation. Then the second day is focused on researching three different computing innovations and discussing these innovations with team members.

Purpose

In this unit students will cover information related to computing innovations needed for the Curricular Requirement for the AP Computer Science Principles exam. This lesson sets the stage for the simulation and has them begin their research.

Standards

Full Course Alignment

CSP Conceptual Framework

- **CRD-1** - Incorporating multiple perspectives through collaboration improves the computing innovations being developed.
- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.
- **IOC-1** - While computing innovations are typically designed to achieve a specific purpose, they may have unintended consequences.
- **IOC-2** - The use of computing innovations may involve risks to your personal safety and identity.

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

Objectives

Students will be able to:

- Identify a computing innovation
- Research computing innovations through the lens of one beneficiary
- Understand how to conduct research on a computing innovation

Preparation

- Prepare the role badges
- Review the Innovation Simulation Planning Guide
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking out the Virtual Modifications for **Day 1**) and **Day 2**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Innovation Simulation Badges** - Handout
- **CSP Innovation Simulation Badges - Editable** - Resource [▼ Make a Copy](#)
- **CSP Innovation Simulation Character Bios** - Handout [▼ Make a Copy](#)
- **CSP Innovation Simulation Nameplates** - Handout
- **CSP Innovation Simulation Project Overview** - Resource

Agenda

The Pre-Unit Pulse

Day 1 (45 minutes)

Warm Up

Activity

Wrap Up

Assessment: Check For Understanding

Day 2 (45 minutes)

Warm Up

Activity

Wrap Up

▼ Make a Copy

- **Cybersecurity and Global Impacts (Unit Slides)** - Slides ▼ Make a Copy
- **Innovation Simulation Part 1** - Slides ▼ Make a Copy

For the students

- **CSP Innovation Simulation Project Guide** - Activity Guide ▼ Make a Copy
- **Computer Science is Changing Everything** - Video (Download)
- **Unit 8 Journal (Digital)** - Resource ▼ Make a Copy
- **Unit 8 Journal (Interactive Notebook)** - Resource ▼ Make a Copy

Vocabulary

- **Computing Innovation** - includes a program as an integral part of its function. Can be physical (e.g. self-driving car), non-physical computing software (e.g. picture editing software), or non-physical computing concepts (e.g., e-commerce).

Teaching Guide

The Pre-Unit Pulse

Have your students complete these questions independently some time before starting this unit.

☰ 1-3

The Pre-Unit Pulse

1

2

3

💡 Teaching Tip

Pre-Unit Pulse Questions: The answers to these questions can provide insights into the preferences, strengths, and motivations of your students. If you'd like to adapt the questions or add in your own, you may want to make a copy of [this Google Form](#) which already has the existing questions populated.

For more tips and ideas of how to use these questions, check out the [CSP Guide to Pre-Unit Pulse Questions](#)

Day 1 (45 minutes)

Warm Up (5 minutes)

Distribute: Give each student a copy of the Unit 8 Journal ([Digital/Interactive Notebook](#)) or have students set up this unit's section in their CSP journals.

Start the Simulation

Teaching Tip

Journaling: Journaling can take many different forms, but in general, it's a tool for individual processing and reflection in a form that can be revisited as students develop their skills and understandings. The medium used for journaling can vary depending on classroom needs.

For more guidance and examples, check out the [CSP Guide to Journaling](#)

Remarks

Welcome to the Future School Convention. You have been chosen to attend this convention because you represent an important stakeholder in the broader school community. Together with a small group you will brainstorm and research computing innovations that you believe will have a positive impact on a the Future School. At the end of the convention, your team will pitch your best idea to a panel of judges.

Group: Place students in groups of 5-6.

Teaching Tip

Small Classes: If you have less than ten students, you can leave out the *PARENT* role and divide into groups of 4-5.

Large Classes: More than one student can have the same character assignment, as long as they are not in the same group. You may also opt to use the blank badges and nameplates to create your own characters. Consider the school counselor or the bus driver.

Distribute: Hand out a random role card to each group. Each group should have at least one *TEACHER*, *STUDENT*, *STAFF*, *PARENT*, and *ADMIN*. Students will need

- A character bio from [CSP Innovation Simulation Character Bios](#)
- A nameplate from [CSP Innovation Simulation Nameplates](#)
- A badge from [CSP Innovation Simulation Badges](#)

Do This: Each students should tape their badge to their shirt and place their nameplate on their desk after folding on the marked lines. When the nameplates and badges are out, students should be in simulation mode.

Discussion Goal: The introductory activity sets the stage for the simulation project, which will run throughout the unit. Encourage students to stay in their roles and speak from their specific points of view. For example, when talking students might say "As the art teacher, I think..." or "As a parent, I know what is best for my student and that is..."

Do This: Read over your role sheet and introduce yourself to your group.

Teaching Tip

Your role involves being the ultimate "role keeper". If you buy-in to the simulation, your students will have a

greater chance of fully participating and keeping engagement high.

Activity (35 minutes)

Remarks

Computing Innovations are created by people to solve problems. They include programs as part of how they function. They can be physical (e.g., self-driving car), non-physical computing software (e.g., picture editing software), or non-physical computing concepts (e.g., e-commerce). The way lives are lived has changed because of computing innovations. Let's rewatch a video to remind ourselves of this fact.

 **Video:** Play [Computer Science is Changing Everything](#). (End at 4:40)

Innovation Simulation Project

 **Brainstorm (10 mins):** Teams spend five minutes brainstorming computing innovations they believe would be beneficial to the Future School. Students should be considering if the innovations include programs as an important part of their function. If not, it's possible that it is not a computing innovation. Remind students to stay in their roles.

Distribute: [CSP Innovation Simulation Project Guide](#)

 **Step 1 - Reflect (10 mins):** Students use this time to consider the wants and needs of their characters and what types of computing innovations would and would not appeal to the character they are playing. Students write their reflection in the Project Guide.

Remarks

Before your teams get started investigating computing innovations for Future School, we want to make sure everyone understands how to conduct the necessary research. Let's examine an innovation that is not related to schools. Let's consider an innovation that would benefit long distance travel.

 **Step 2 - Research (10 mins):** Model doing a Google search with your students to determine a computing innovation related to long distance travel. You might directly Google the term, search out articles, or turn to Wikipedia. After you have found a computing innovation that is promising, model filling out the table in the Project Guide. You can project a copy of the Project Guide while doing this, flipping back and forth between the guide and the website you are using. You do not need to fill out the last section in the table, but you should discuss with your students how they will fill this section out. It is important to note that students will have the next lesson to conduct their own research. Today, your goal is to only model how to approach this research and using the Project Guide.

Teaching Tip

Doing this research may take some practice. We recommend trying a few different search terms a few times before class on your own, to assess where things may be tricky. Some example terms to search might be general terms like "New technologies for schools" or "Technologies to change schools". You could also model searching for technology related to specific roles such as searching for "Technology for art teachers".

It's important to model to students how to use search terms to narrow down to a computing innovation of interest.

Wrap Up (5 minutes)

Remarks

Thank you delegates for beginning the process of identifying computing innovations for The Future School. I'm looking forward to hearing your proposals soon.

As a team, you will need to work effectively together, evaluating and considering these computing innovations. Effective teams work on communication, consensus building, conflict resolution, and negotiation. You will address these skills throughout the project!

 **Do This:** End the day by instructing students to put away their badges and nameplates. You may want to collect them so they are not lost.

 **Discuss:** *What is the purpose of a computing innovation?*

Discussion Goal: A computing innovation is designed to solve a problem.

Journal: (Click through animation to see definition). Students add to their journal the definition for computing innovation.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: How has your life benefited from a computing innovation? **Note:** You are out of the simulation for this question! Speak from your own personal experiences.

 4



Check For Understanding - Day 1

Day 2 (45 minutes)

Warm Up (5 minutes)

 **Distribute:** Pass out badges and nameplates from the previous lesson. While you are doing this, draw students' attention to the red box in the slide header that indicates the class is in simulation mode. From this point on in the class, students are thinking from the perspective of their given character.

Group: Instruct students to sit with their team from the previous lesson.

Remarks

Welcome back to day two of the Future School Convention. I hope you enjoyed your first day getting to know your fellow participants. Today the real work begins - you will be researching the computing innovations that you believe will benefit the Future School. I'm sure you have lots of ideas. Let's get started!

 **Discuss:** *Why do you believe the Future School is in need of computing innovations? What can computing innovations provide to a community?*

Discussion Goal: In the near future, we will discuss the harms that computing innovations can bring to society. For today, we want to focus on the positive benefits.

Computing Innovations can:

- help schools run more efficiently

- help certain groups gain a voice and be heard
- make learning more fun!
- save time and money

Activity (35 minutes)

 **Research:** Students research three different innovations that they believe their character would recommend for the Future School.

In their Project Guides they mark down:

- The name of the innovation
- Sources
- What data is being collected
- Brief descriptions of the function and purpose of the innovation
- Why their character would recommend it

Students should regularly check in with their group to make sure no one else is researching the same innovation. For each group, there should be a total of 15 (or three times the amount of group members) innovations researched.

Wrap Up (5 minutes)

 **Discuss:** *Which of your computing innovations would bring the most positive change to the school community? Share with your group. *

With your group and then with the whole class discuss the common characteristics of these computing innovations that would bring the most positive change.

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Considering your character's perspective, which computing ideas of your team members seem the best? Which ones are you concerned about?



Check For Understanding - Day 2



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